"Seasonal compound renewable energy droughts in the Unites States"

We thank the editor and the reviewers for the detailed comments. Responses are below.

Referee: 1

COMMENTS TO THE AUTHOR(S)
This paper analyzes the frequency, duration, and intensity of compound renewable generation (wind, solar, and hydropower) droughts at seasonal timescales across the U.S. and demonstrates the implications of these generation droughts from the historical record on resource adequacy over the WECC. This study differs from the established literature on compound renewable droughts by 1) primarily focusing on long-term droughts (beyond the ~hours to days timescales of most grid studies) and 2) including more detailed representation of hydropower in the compound drought calculations as a variable renewable resource. My comments are as follows:

1. Please clarify – when defining VRE drought as below the 40th percentile for each generation type, does this mean VRE drought is when the combined wind/solar/hydropower generation is below the 40th percentile for the combined generation of all three resources? Or alternatively does this mean when the combined generation is below the 40th percentile for any of the three resources? I assume it is the former, but the authors should be more explicit.

The 40th percentile threshold requires each generation source independently to be lower than the threshold to qualify as a compound event. This has been clarified in the text, and an equation has been added to for further clarity.

2. How does the 40th percentile definition for compound VRE drought compare with the metrics used by BAs/utilities to assess resource adequacy? Put another way – in planning for resource adequacy, do BAs/utilities use more conservative or more aggressive criteria than that used in this study?

In practice BAs will likely use an impact-based threshold that is specific to their BA and reflects the proportion of renewables relative to other generation sources. This would be an interesting extension to explore but is beyond the scope of this paper.

3. The analysis of the historical record of compound droughts is interesting to me and well presented. I find this to be a very useful context for the paper's other results. The authors should discuss, however, whether they expect any of these characteristics from the historical record to change as climate change effects set in during future years, even though forcing the models here with GCMs is beyond the study scope.

We completely agree that a study of the future changes in energy droughts is an interesting and important extension. We have another paper in review that directly looks at this question (https://eartharxiv.org/repository/view/8377/). Future droughts are a more complicated due to the need to include multiple future projections of climate but also the need to project future infrastructure development. In general, we expect the characteristics of drought to change in the future, but the growth of wind and solar infrastructure can have a mitigating effect on the severity of drought. When more wind and solar plants exist within a BA, the weather conditions required to put all the plants in drought are stricter, and less likely to occur.

4. The description of the drought events in the grid simulations is somewhat vague and I didn't find a more specific description of what parameters defined each scenario in the SI. The authors should provide an explicit description of Events 1-5 for completeness.

Thanks for the comment. We have added Table 2 to the paper which details each event, the BAs affected and the year and month the event occurred.

Table 2: The 5 worst (1 is the worst event) energy droughts affecting Western US BAs from 1982-2019. These 5 events are modeled in the case study.

Event	BA	Year	Month
1	AVA, BPAT, CISO, IPCO, NEVP, PACW, PSEI, SRP	2000	10, 11
2	AVA, BPAT, CISO, IPCO, NEVP, PACW, PGE, SRP, WALC	2019	10, 11
3	BPAT, CISO, NEVP, NWMT, PACW PSCO, PSEI, SWPP, WALC	2000, 2001	10
4	AVA, CISO, IPCO, NEVP,	1985	10, 12
	PSEI, SRP, WALC	1985	10, 12
5	CISO, NEVP, SWPP	1986	11, 12

5. Overall, I do feel like this paper presents many interesting insights about the characteristics of compound renewable droughts and starting insight into how this affects resource adequacy, but I think the paper needs some more discussion about the implications of these characteristics for longer-term planning of electricity systems. As wind and solar penetration increases and the significance of these droughts become more important for planning to ensure resource adequacy, the authors should speak to what the characteristics of these droughts mean for what steps should be taken by BAs and utilities to prepare for or counteract the negative consequences of these droughts. For example, does the longer-term compound VRE droughts from the historical record justify investments in seasonal energy storage resources or firm zero-carbon generation such as geothermal? Do the presented characteristics of these compound VRE droughts justify other investments such as

transmission targeted to connect regions experiencing compound VRE drought with other regions that can support it? I understand that these types of questions are beyond the explicit scope of this study, but in order to have more significance for the energy system planning literature and practice, I think these should be further discussed.

Thanks for this comment. We have added a paragraph to the discussion to address this:

"These results have implications for the long-term grid planning, especially in a decarbonized or low carbon future grid. Current storage technology and peaking capability may not be sufficient to mitigate droughts which last many months in duration. As renewable buildout continues the need for long duration energy storage will likely increase if we are to ensure future demands are met. To evaluate potential impacts, individual BAs should assess compound drought characteristics across multiple scales. For example, if low production periods within a seasonal drought occur relatively uniformly vs concentrated during smaller periods of the drought, could strongly influence the amount and type of energy storage required. Mitigating strategies for energy droughts such as investment in storage, transmission, or other generation sources will be regionally specific and need to be tailored by each BA to meet their unique needs. Such strategies should be developed with a comprehensive understanding of local climate patterns, electricity generation and load profiles, and other factors that affect grid reliability,"

After addressing these comments, the paper can be considered for publication.

Referee: 2

COMMENTS TO THE AUTHOR(S)
Review of "Seasonal compound renewable energy droughts in the United States" by Bracken et. al. for Environmental Research: Energy. In this manuscript, the authors analyze seasonal compound variable renewable energy (VRE) droughts across the contiguous United States at the balancing authority level. Overall, this manuscript would be a beneficial addition to the energy drought literature and in steps towards its inclusion in more established grid operations.

Major Comments
1. Why is the analysis based on a dataset ending in 2019, given that we are currently in 2025. Is it a limitation of the TGW climate simulation dataset? Additionally adding more description, in addition to the citations, should be added in the supplemental materials esp. about computing wind and solar generation from the underlying solar radiation and wind speeds. Was the temperature dependence included in solar generation estimation?

Yes, this is a limitation of the TGW data, if the data is updated in the future, we will update the underlying renewable datasets.

More detail has been added to the supplemental material regarding the wind and solar data generation. Temperature dependence was included in the model based on the manel type as a derating factor, where higher temperature leads to lower panel efficiency.

2. One of the main selling points of this manuscript is analysis at the balancing authority level, given that BAs represents the grid-relevant decision-making scale. I totally agree with them on this! My concern is with the inclusion of the smaller BA's e.g., PacificCorp West, Portland General Electric, L.A. Dept of Water and Power) in the analysis. My guess is that these smaller BA's are biasing their results esp. in the cluster analysis (A lot of smaller BAs in the west compared to a few large BAs in the east). It might be helpful to include a cut-off for the renewable energy capacity in the BA's. Additionally adding a column representing total renewable energy capacity in BA in table 1 would be useful.

Considering the relative geographical size of a BA and renewable capacity is a very valid consideration. This is something we plan on looking into in more detail in the future. In addition, we treat the relative contribution of each resource as contributing equally to a resource drought which is also a place for improvement. While these smaller BAs should not be treated equally in terms of their overall grid impact, in most cases the results are not biased because they assess each BA independently or regionally in the case of the climate analysis. The only case where all BAs are lumped together is the duration histogram. To show the results are substantially similar with small BAs included we excluded any BA smaller than PACW (~3 GW of renewables). The figure below shows the updated duration histogram is

3. What is the rationale for the 40th percentile threshold value? How sensitive are the results to this threshold when compared to 35th or 45th percentile? Additionally, the description in line 148 and the method for the calculation of a compound VRE drought can come off as a bit vague. Adding an equation as was done for severity covariance might be useful.

The threshold in this type of analysis is arbitrary. We chose the 40th percentile for all BAs to be as low as possible to capture the most extreme events while also representing enough events in every BA to conduct the analysis. In a BA-specific application the threshold should be chosen based on the potential impacts and could be chosen differently for each generating resource.

Figures 2 and 3 show the drought duration and magnitude with a 40th and 35th percentile thresholds respectively. In general, the lower threshold captures fewer events in fewer BAs where the overall duration of events is shorter. For example, there are no 5 month events when using the 35th percentile threshold, and there are no events captured in the TVA BA.

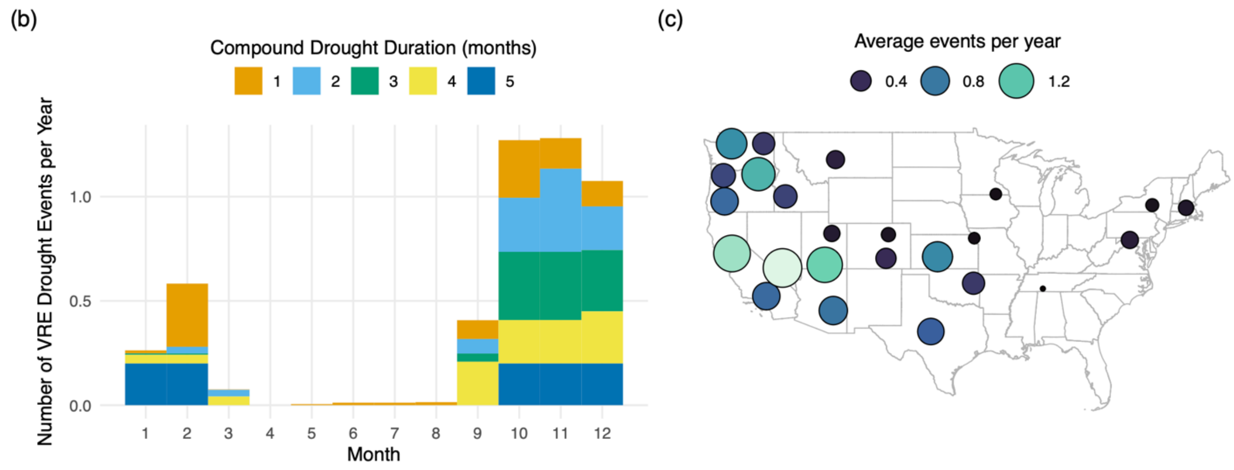

Figure 1: Drought duration and magnitude with a 40th percentile threshold

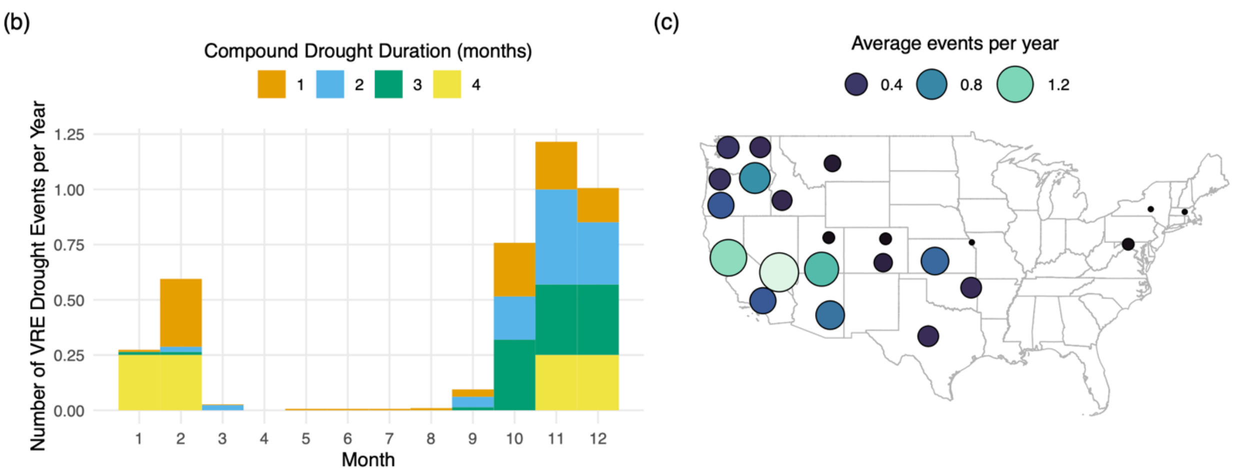

Figure 2: Drought duration and frequency with a 35th percentile threshold

We have completed a more comprehensive sensitivity in a previous paper (https://www.sciencedirect.com/science/article/pii/S0960148123014659) involving shorter drought events which may be of interest: https://ars.els-cdn.com/content/image/1-s2.0-S0960148123014659-mmc1.pdf
In general, a lower threshold decreases the number of events identified, increases the average drought length, and decreases the average drought magnitude.

We have also added Equation 1 which more accurately describes how a drought is defined on a particular month.

4. In Section 5, the authors use strong language to characterize their results. In particular, the phrase "clear causality" in line 378 and line 383 is particularly strong

and does not reflect the results shown. Instead of causality shouldn't it be statistical significance? Furthermore, these results establish the potential for predictability rather than predictability itself. Furthermore, do the colors and color legend in Figure 5 represent the p-values or the correlation values. I am asking since the legend only goes from -0.1 to 0.1.

Thank you for the comment. We have modified the language to say, "statistically significant connections" instead of "clear causality". In Figure 5, the color bar represents correlation. We recognize that these are low values, but they do still represent a statistically significant relationship that could form the bases for a prediction model, which is why we have included them.

5. Additional description and setting up of the WECC 2030 ADS is needed. Furthermore, it might be useful to show the whole distribution of the LMPs, given that such droughts or shortages are likely to cause spikes.

Thank you for your comment. We have added further description of the WECC ADS.

"The WECC 2030 ADS is the base nodal dataset for the Western Interconnection baseline analysis (WECC, 2024). The WECC 2030 ADS is a comprehensive scenario, incorporating input from all planning regions. It aims to project the Western Interconnection's infrastructure for the year 2030, based on <u>the best available knowledge at the time of the case's release as well as existing state and federal laws</u>. The transmission network topology for the WECC 2030 ADS Production Cost Model was carried over from a previous WECC study—the 2030HS (Heavy Summer) Power Flow dataset, which was compiled by the WECC Reliability Assessment Committee using GE PSLF software. The transmission topology was imported into the 2030 ADS Production Cost Model case as the basis for the transmission network topology and represents the best available projection of anticipated new generation, generation retirements, transmission assets, and load growth in the 10-year planning horizon within the WECC grid planning community."

We have shown the distribution of LMPs for the baseline scenario in the below figure (Figure 3). LMPs exhibit significant spatial and temporal variability, especially among extreme values. This makes it difficult to extract meaning from the exact values of LMPs. For this reason, we have elected to represent the impact of droughts in the manuscript through the annual average change compared to the baseline scenario, by zone.

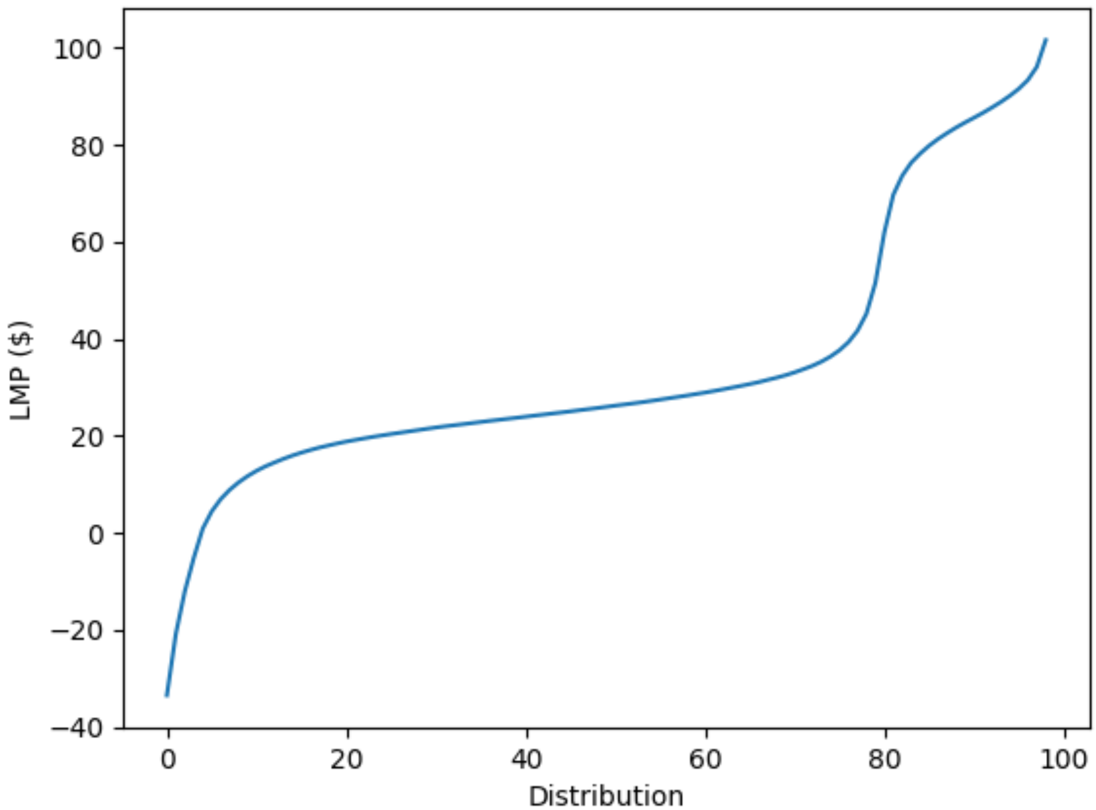

Figure 3: Distribution of LMPs in the baseline scenario. LMPs exhibit significant spatial and temporal variability, with strong peaks both in positive and negative LMPs.

Minor Comments

1. Do the authors mean Table "1" in line 147?

Yes, thanks. This has been corrected.

2. How were the AO, NAO and ENSO indices calculated and where were the values taken from? Additional descriptions and citations are needed

Thank you for the question. We did not calculate the AO, NAO, and ENSO indices ourselves but obtained them from the official NOAA website (https://psl.noaa.gov/data/climateindices/list/). These indices were then used to assess their correlation with energy droughts in our study. We have now provided additional details on the data sources and included the appropriate citation in the revised manuscript.

3. Citations are necessary for the following

a. Citation needed for the sentence ending in Line 56 on low wind speeds in 2015.

We have added the citation:
D. Rife, N. Y. Krakauer, D. S. Cohan, J. C. Collier, A new kind of drought: Us record low windiness in 2015, Earthzine (2016).

b. Nodal Production Cost Model (Line 89). Additionally, a further description of nodal PCM's and the difference with unit commitment/economic dispatch models, capacity expansion and other models would be beneficial to the users.

Thank you for your comment. The nodal PCM simulates grid operation using unit commitment/economic dispatch. Capacity expansion models are not utilized in this study, but the critical difference is that capacity expansion models simulate investment in new capacity in addition to operations. We have clarified this language in the text.

c. Citation needed in line 466 for the inclusion of "stress periods" in capacity expansion models.

Added a reference to Anderson 2024:

O. Anderson, C. Bracken, C. D. Burleyson, A. Pusch, N. Yu, Improved decarbonization planning through climate resiliency modeling, IEEE Access 12 (2024) 128494–128508. doi:10.1109/access.2024.3451957

4. Figure 6 (bottom plot), is the change in the LMP's in terms or ratio or actual dollar amounts? In the text the authors describe it in terms of ratios.

Thank you for the comment. We have edited the language in Figure 6 to reflect that all values are ratios, rather than actual dollar amounts.